Methodology Passport Card

1. Project Information

Field	Value
Title	Multidimensional Spacetime and the Theory of Relativity: A Computational Benchmark of Black-Hole Properties in D-Dimensional Spacetime
Authors	Talgat Bakytuly, Dauletuly Amirlan
GitHub repository	https://github.com/Tako1337/spacetime-dimensions-bh

2. Research Type

Mixed-Methods.

- Qualitative part (RQ1, RQ2): literature synthesis. These questions are descriptive and need no measurements (see Task 1).
- Quantitative part (RQ3): computational experiment based on exact analytic solutions (Tangherlini, Myers-Perry).

Justification: there are no direct observations of extra dimensions, so RQ3 is tested by computing theoretical models under fixed conditions.

3. Hypotheses

- **H0:** for every D from 5 to 11, black-hole properties do not differ qualitatively from D = 4: the solution has an upper bound on spin (the horizon disappears for $\chi \le \chi_{cap}$) and stable circular orbits exist.
- **H1:** for at least one D ≥ 5, the properties differ qualitatively from D = 4: a horizon exists at $\chi = \chi_{cap} = 4$, whereas in D = 4 it disappears for $\chi > 0.5$.

H0 is rejected if B(D) = 1 for at least one D ≥ 5 while B(4) = 0, and all guardrail metrics pass.

4. Variable Matrix

Independent variables

Variable	Values
Spacetime dimension D	4, 5, 6, 7, 8, 9, 10, 11
Spin $\chi = a / \mu^{(1/(D-3))}$	41 values on [0, 4]
Solution family	Schwarzschild-Tangherlini; Myers-Perry (single spin)

Dependent variables

Variable	Units
Horizon existence	0/1
Horizon radius r_h	G = c = 1, M = 1
Ratio r_{ph} / r_h	dimensionless
Stable circular orbits, r_{ISCO}	bool / length
Horizon area A, entropy S = A/4	G = c = ħ = 1
Hawking temperature T_H	G = c = ħ = k_B = 1
Maximum spin χ_{max}	dimensionless
Latency per cell	ms (median of 30 runs)
Peak memory	MB

Controlled variables

Factor	Value
Units	G = c = ħ = k_B = 1
Mass	M = 1

Number of nonzero angular momenta	1
Charge, Λ, matter fields	none (vacuum)
Software	Python 3.13, NumPy 2.5.3, SciPy 1.18.1, PyYAML 6.0.3
Root-finding tolerance	xtol = rtol = 1e-14 (`brentq`)
Spin grid	linspace(0, 4, 41)
Repeats / warm-up	30 / 3
Environment	Docker python:3.13-slim, exact dependency pins
Input data	deterministic grid from config.yaml

5. Metrics and Baseline

Primary metric

B(D): equals 1 if a horizon exists at χ = χ_cap = 4, otherwise 0. The presence of a spin bound separates D = 4 (χ ≤ 0.5) and D = 5 (χ < 1) from D ≥ 6, where no bound exists. This directly tests RQ3.

Secondary metrics

r_ph / r_h; stable circular orbits; T_H; entropy; χ_max; latency; peak memory.

Guardrail metrics

Metric	Threshold
Root residual of the horizon equation	≤ 1e-10
Smarr relation error (D−3)M = (D−2)(TS + ΩJ)	≤ 1e-9
D = 4 reproduction (r_h = 2M, Kerr bound a = M)	tolerance 1e-9
Computation time per cell	≤ 1000 ms

If any threshold is violated, the conclusion on the hypothesis is invalid and the script returns a nonzero exit code.

Baseline

D = 4: Schwarzschild (χ = 0) and Kerr (χ ≤ 0.5) solutions, i.e. standard general relativity. Checked in unit tests: r_h = 2M, r_ISCO = 6M, r_ph/r_h = 3/2, T = 1/(8πM).

6. Pipeline Test Results

D	B(D)	χ_max	r_ph/r_h	Stable orbits
4	0	0.5	1.500	yes
5	0	1.0	1.414	no
6	1	≥ 4	1.357	no
7	1	≥ 4	1.316	no
8–11	1	≥ 4	1.285–1.223	no

All guardrails pass. χ_max for D = 4 and 5 is determined to the grid step. Full analysis is planned for Week 7.

7. Limitations

- The experiment checks known analytic results and gives no observational evidence for extra dimensions.
- Stability of ultraspinning horizons (D ≥ 6) is not tested.
- Only vacuum solutions with a single angular momentum are considered; black rings and other horizon topologies are covered in the literature part only.